

 TECHNICAL QUICK-START GUIDE

Local Installation & Setup Guide (From Scratch)

Complete step-by-step guide to configure, run, and test the Reservation Management System on your local machine (Mac / Windows / Linux).

1 Project Architecture & Core Components

FRONTEND

Angular 19

Port: 4200 or 8080

BACKEND API

ASP.NET Core 9

Port: 5000 / Swagger UI

DATABASE

MySQL / MariaDB

Port: 3306 or 3307

DB MANAGER

phpMyAdmin

Port: 8081

2 System Prerequisites

Required Tool	Recommended Version	Role
Docker & Docker Compose	Latest (Docker Desktop)	For 1-Click automated deployment (Recommended)
Node.js & npm	Node v20.x LTS + npm 10.x	To build & run the Angular Frontend locally
.NET SDK	.NET 9.0 SDK	To build & run the C# Backend Web API locally
MySQL / MariaDB	MySQL 8.0+ / MariaDB 10.11+	Local database server (if running without Docker)

3 Method 1: Automated Start with Docker (1-Click - Recommended)

⚡ Full Stack Launch in 1 Single Command

FASTEST & EASIEST

This method automatically downloads, builds, and runs the MariaDB Database, .NET 9 Web API, Angular Frontend (Nginx), and phpMyAdmin with zero manual database configuration.

1 Open terminal in the project root directory:

```
# Navigate to the project root
cd /path/to/ReservationSystem
```

2 Launch all containers with Docker Compose:

```
docker-compose up -d --build
```

✅ Done! Docker will automatically create the database, import the script.sql schema, seed the default Admin account, and serve the application.

3 Access your services in the browser:

Service	Local URL	Default Credentials
🌐 Web Application	http://localhost:8080	admin / admin123
⚙️ Backend API	http://localhost:5000	Endpoints /api/*
📁 phpMyAdmin	http://localhost:8081	Server: db User: root Pass: rootpassword

4 Useful Docker commands:

```
# Check container status
docker ps

# View live container logs
docker-compose logs -f

# Stop and remove all containers
docker-compose down
```

4 Method 2: Manual Setup Without Docker (Development Mode)



Local Development (Live Hot-Reload & Debugging)

DEVELOPER MODE

A Step 1: Setup Local Database (MySQL / MariaDB)

1. Start your local MySQL service (via XAMPP, Homebrew, or native MySQL server).
2. Open phpMyAdmin or your MySQL client (e.g., DBeaver, TablePlus, Workbench).
3. Create a new database named: `ReservationSystemDb`.
4. Import the SQL initialization file located at: `Server/script.sql`.

B Step 2: Start the Backend (.NET 9 Web API)

Verify your database connection string in `Server/appsettings.json`:

```
"DefaultConnection":  
"Server=localhost;Port=3306;Database=ReservationSystemDb;User=root;Password="
```

Open a **first terminal tab** and run the following commands:

```
# 1. Navigate to Server folder  
cd Server  
  
# 2. Restore NuGet dependencies  
dotnet restore  
  
# 3. Build and launch the Backend API  
dotnet run
```

👉 The Backend will run on **`http://localhost:5000`**. Interactive Swagger API documentation is available at **`http://localhost:5000/swagger`**.

C Step 3: Start the Frontend (Angular 19)

Open a **second terminal tab** and run the following commands:

```
# 1. Navigate to App folder  
cd App  
  
# 2. Install npm packages (with legacy-peer-deps flag)  
npm install --legacy-peer-deps  
  
# 3. Start Angular live development server  
npm start
```

👉 The Frontend will open on **`http://localhost:4200`** with automatic live reload on every source code edit.

5 Default Authentication Credentials



Auto-Seeded Administrator Account

The system automatically generates and seeds this initial administrator account on database startup:

USERNAME
admin

PASSWORD
admin123

6 Troubleshooting & FAQ

Issue Description	Probable Cause & Solution
Port 3306 conflict Error: bind: address already in use	A local MySQL service is already bound to port 3306. Fix: The project's <code>docker-compose.yml</code> maps the database to port 3307:3306 to prevent any port collisions.
npm install dependency error ERESOLVE unable to resolve dependency tree	Minor peer dependency version mismatch in Angular packages. Fix: Always use the legacy flag: <code>npm install --legacy-peer-deps</code> .
Blank Screen on first launch at <code>http://localhost:8080</code>	The backend container is still applying initial database migrations. Fix: Wait 10 seconds and perform a Hard Refresh (Cmd+Shift+R or Ctrl+F5).
Forcing a clean rebuild	To force Docker to recompile modified code from scratch: <code>docker-compose down && docker-compose up -d --build</code>